OSPF Questions 2

8 questions | Original English questions and source answers

Question 1

Refer to the exhibit. Which configuration issue is preventing the OSPF neighbor relationship from being established between the two routers?

```
R2#show running-config
Building configuration...
!
interface GigabitEthernet2/0
ip address 192.168.0.2 255.255.255.252
negotiation auto
!
router ospf 1
router-id 2.2.2.2
passive-interface default
no passive-interface GigabitEthernet2/0
network 192.168.0.2 0.0.0.0 area 0
!
R1#show running-config
Building configuration...
!
interface GigabitEthernet1/0
mtu 1600
ip address 192.168.0.1 255.255.255.252
negotiation auto
!
router ospf 1
router-id 1.1.1.1
passive-interface default
no passive-interface GigabitEthernet1/0
network 192.168.0.1 0.0.0.0 area 0
!
```



Options

A. R1 has an incorrect network command for interface Gi1/0

B. R2 is using the passive-interface default command

C. R2 should have its network command in area 1

D. R1 interface Gi1/0 has a larger MTU size

Answer

D. R1 interface Gi1/0 has a larger MTU size

Source: CCNA Training » OSPF Quiz 2.pdf | Question 1

Question 2

Refer to the exhibit. What is the next hop address for traffic that is destined to host 10.0.1.5?

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

Gateway of last resort is not set

C       1.0.0.0/8 is directly connected, Loopback0
        10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O         10.0.1.3/32 [110/100] via 10.0.1.3, 00:40:07, Serial0
C         10.0.1.0/24 is directly connected, Serial0
O         10.0.1.5/32 [110/5] via 10.0.1.50, 00:40:07, Serial0
O         10.0.1.4/32 [110/10] via 10.0.1.4, 00:40:07, Serial0
```

Options

A. 10.0.1.50

B. 10.0.1.3

C. Loopback 0

D. 10.0.1.4

Answer

A. 10.0.1.50

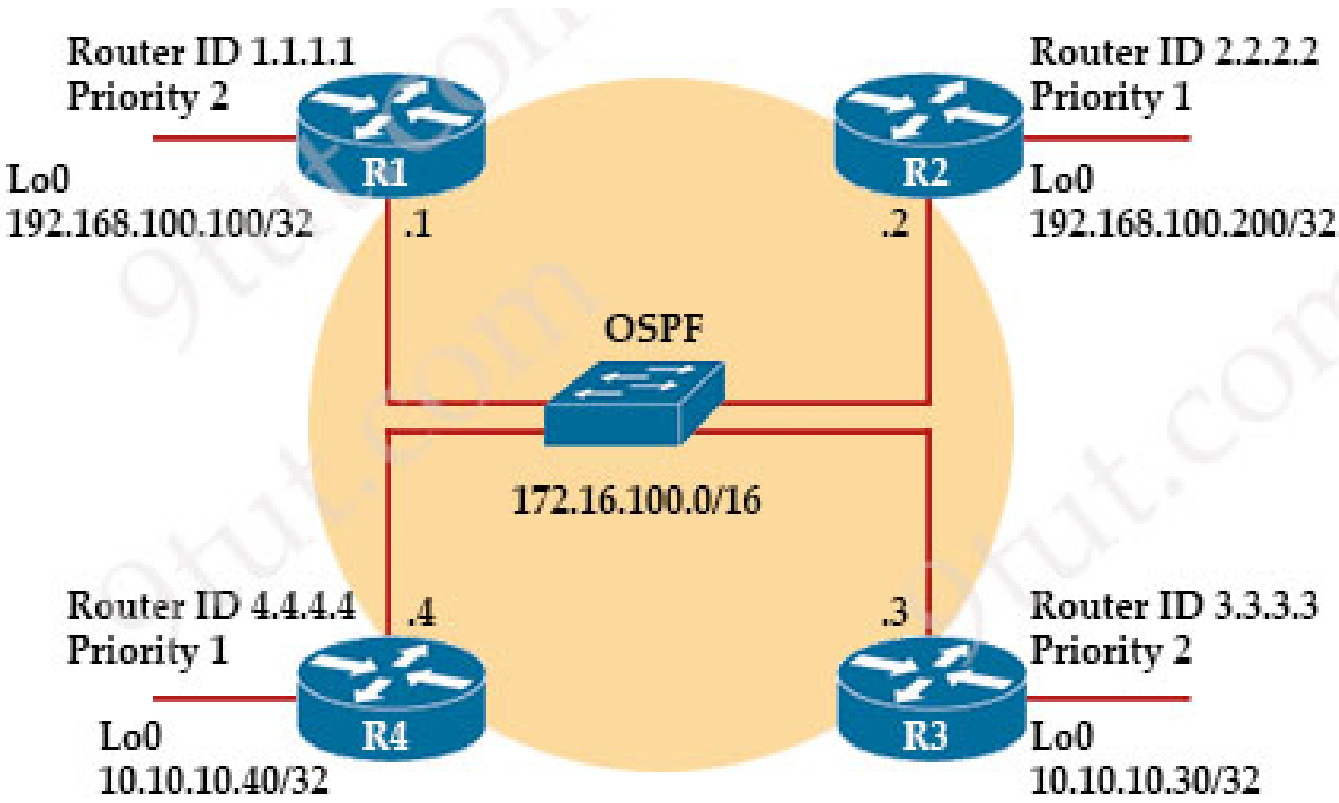
Source explanation

Traffic destined to host 10.0.1.5 will match the entry “O 10.0.1.5/32 …” so the next-hop address will be 10.0.1.50.

Source: CCNA Training » OSPF Quiz 2.pdf | Question 2

Question 3

Refer to the exhibit. If the switch reboots and all routers have to re-establish OSPF adjacencies, which routers will become the new DR and BDR?



Options

A. Router R3 will become the DR and router R1 will become the BDR.

B. Router R4 will become the DR and router R3 will become the BDR.

C. Router R1 will become the DR and router R2 will become the BDR.

D. Router R3 will become the DR and router R2 will become the BDR.

Answer

A. Router R3 will become the DR and router R1 will become the BDR.

Source explanation

After the new election, R3 and R1 have highest priority (of 2) so they will be elected DR and BDR. R3 will be elected DR because its has higher router ID (of 3.3.3.3).

Source: CCNA Training » OSPF Quiz 2.pdf | Question 3

Question 4

Which step in the link-state routing process is described by a router sending Hello packets out all of the OSPF-enabled interfaces?

Options

A. electing the designated router

B. establishing neighbor adjacencies

C. injecting the default route

D. exchanging link-state advertisements

Answer

B. establishing neighbor adjacencies

Source: CCNA Training » OSPF Quiz 2.pdf | Question 4

Question 5

Which two minimum parameters must be configured on an active interface to enable OSPFv2 to operate? (Choose two)

Options

A. OSPF stub flag

B. OSPF process ID

C. OSPF area

D. OSPF MD5 authentication key

E. IPv6 address

Answer

B. OSPF process ID

C. OSPF area

Source explanation

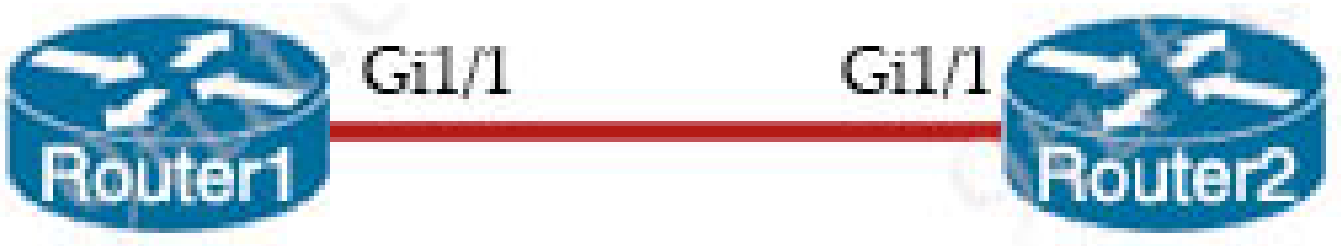
To configure OSPFv2 under interface mode we must use the command “ip ospf {process-id} area {area-id}”.
For example:
R1(config-if)#ip ospf 1 area 2

Source: CCNA Training » OSPF Quiz 2.pdf | Question 5

Question 6

Refer to the exhibit. Refer to the exhibit. After the configuration is applied, the two routers fail to establish an OSPF neighbor relationship. what is the reason for the problem?

```
Router1(config)#interface GigabitEthernet1/1
Router1(config-if)#description ***Connection to
Router2***
Router1(config-if)#ip address 10.10.10.1
255.255.255.252
Router1(config-if)#ip ospf hello-interval 5
Router1(config)#router ospf 1000
Router1(config-router)#router-id 1.1.1.1
Router1(config-router)#network 10.10.10.0 0.0.0.3
area 0
Router2(config)#interface GigabitEthernet1/1
Router2(config-if)#description ***Connection to
Router1***
Router2(config-if)#ip address 10.10.10.2
255.255.255.252
Router2(config)#router ospf 1001
Router2(config-router)#router-id 2.2.2.2
Router2(config-router)#network 10.10.10.0 0.0.0.3
area 0
Router2(config-router)#passive-interface default
Router2(config-router)#no passive-interface
GigabitEthernet1/1
```



Options

A. The OSPF router IDs are mismatched

B. Router2 is using the default hello timer

C. The network statement on Router1 is misconfigured

D. The OSPF process IDs are mismatched

Answer

B. Router2 is using the default hello timer

Source: CCNA Training » OSPF Quiz 2.pdf | Question 6

Question 7

An engineer must configure an OSPF neighbor relationship between router R1 and R3. The authentication configuration has been configured and the connecting interfaces are in the same 192.168.1.0/30 subnet. What are the next two steps to complete the configuration? (Choose two)

Options

A. configure the hello and dead timers to match on both sides

B. configure the interfaces as OSPF active on both sides

C. configure the same router ID on both routing processes

D. configure both interfaces with the same area ID

E. configure the same process ID for the router OSPF process

Answer

B. configure the interfaces as OSPF active on both sides

D. configure both interfaces with the same area ID

Source explanation

We don’t need to configure hello and dead timers for OSPF neighbor as OSPF can auto negotiate the network type (with the same hello and dead timers on both sides). We just have to make sure both interfaces are running
OSPF and they are in the same area ID.

Source: CCNA Training » OSPF Quiz 2.pdf | Question 7

Question 8

Refer to the exhibit. What action establishes the OSPF neighbor relationship without forming an adjacency?



```
R1#show ip ospf int gi0/0
Gig0/0/0 is up, line protocol is up
  Internet Address 10.201.24.8/28, Area 1, Attached via Network Statement
  Process ID 100, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID   Cost   Disabled   Shutdown   Topology Name
        0           1       no         no          Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.1.1, Interface address 10.201.24.8
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:05

R2#show ip ospf int gi0/0
Gig0/0/0 is up, line protocol is up
  Internet Address 10.201.24.1/28, Area 1
  Process ID 100, Router ID 172.16.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.16.1.1, Interface address 10.201.24.1
  No backup designated router on this network
  Timer intervals configured, Hello 20, Dead 80, Wait 40, Retransmit 5
```

Options

A. modify priority

B. modify hello interval

C. modify process ID

D. modify network type

Answer

B. modify hello interval

Source explanation

In this question, R1 & R2 will not establish OSPF neighbor relationship because the hello & dead intervals are not the same.
If we modify the hello & dead intervals to the same values then R1 & R2 will become OSPF neighbors. But they will establish OSPF adjacency too.
Answer 'modify process ID' is not correct as the process ID is only locally significant.
Answer 'modify hello interval' is partially correct as we need to change both “hello and dead” interval, not “hello” interval only.
Answer 'modify priority' is a good answer because if we change both the priorities to 0 then they cannot elect DR/BDR so the adjacency cannot be complete.
Answer 'modify network type' is a good answer too because if we change one of the network type to “point to point” then OSPF neighbor relationship can still come up but adjacency cannot (no routes are getting exchanged).
But two good answers are only correct after fixing the hello & dead intervals to the same value (so that OSPF neighbor relationship can come up) so answer 'modify hello interval' should be the best choice here.
Note:
Neighbors
Routers that share a common segment become neighbors on that segment. Neighbors are elected via the Hello
protocol. Hello packets are sent periodically out of each interface using IP multicast (Appendix B). Routers become neighbors as soon as they see themselves listed in the neighbor’s Hello packet.
Two routers will become neighbors if they agree on the following: Same Area ID, same authentication (if used), same hello & dead intervals, same subnet mask and same stub area flag.
Adjacencies
Adjacency is the next step after the neighboring process. Adjacent routers are routers that go beyond the simple
Hello exchange and proceed into the database exchange process.
The adjacency building process takes effect after multiple stages have been fulfilled. Routers that become adjacent will have the exact link-state database. The states an interface passes through before becoming adjacent to another router are: Down -> Attempt (optional) -> Init -> 2-Way -> Exstart -> Exchange -> Loading -> Full.
At Full state the adjacency is complete.
Reference: https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/7039-1.html#t22
-> Two OSPF neighbors do not exchange any routing information – the only packets they exchange is Hello packets.
Adjacencies on Point-to-Point Interfaces
OSPF will always form an adjacency with the neighbor on the other side of a point-to-point interface such as point-to-point serial lines. There is no concept of DR or BDR. The state of the serial interfaces is point to point.

Source: CCNA Training » OSPF Quiz 2.pdf | Question 8